

3. Low-Power Artificial Intelligence for Monitoring Student Well-Being and Burnout Risk in Low-Connectivity Contexts in **Brazil**

Ivo Calado, Rafael Durelli, Marcelo Guimarães, Diego Dias

3.1. Current scenario

Brazilian public schools are struggling with rising levels of student anxiety, chronic stress, and academic exhaustion, phenomena that deteriorate engagement, academic achievement, and school retention (Zhang & Liu, 2025; OECD, 2025a). Two features of the Brazilian system make this especially difficult to address. First, mental health remains heavily stigmatised: a PISA supplementary analysis found that roughly 70% of young people avoid admitting they are stressed (PISA Focus Report, 2022), which means the students most in need of support are also the least likely to disclose it. Second, the institutional capacity to notice and respond is thin, as only 15.7% of Brazilian public schools have a psychologist on site (BRAZIL, 2024; G1, 2025). Therefore, identification and response depend on the informal, unsystematic judgment of already-overloaded teachers.

Poor connectivity itself does not cause the problem; it is, however, a compounding factor. With only 63.7% of schools in Brazil's North region connected to the internet (BRAZIL, 2024; TELETIME, 2025), the cloud-based platforms that dominate the AI in education market are simply not usable in many of the schools where psychosocial need is greatest (Sarker, 2021; Liu et al., 2024). The result is a double exclusion: students in low-connectivity regions face both weaker in-person psychosocial infrastructure and no realistic path to AI-based support tools designed for high-bandwidth contexts.

3.2. Proposed solution

Addressing this gap requires a tool built for the conditions schools actually have, not the conditions AI vendors assume. This work proposes an offline-first mobile application that administers a validated psychosocial screening instrument, the DASS-21 (Depression, Anxiety and Stress Scale), adapted for Brazilian adolescents (Cao et al., 2023). Rather than transmitting responses to a cloud server, the app scores them locally using a lightweight

classification model (TinyML), a form of low-power AI that mirrors the low-connectivity, minimal-hardware design principles at the core of AIED Unplugged (Uema et al., 2026). Students complete the questionnaire on a shared school device; the app classifies each response into a risk band (low, moderate, or severe stress); and school staff receive an aggregated, anonymised alert, for example “five students in this class flagged at moderate-to-severe risk,” rather than individually identifying names to anyone outside a trained counsellor's view.

This design choice is deliberate, not incidental. It replaces a slow, informal, and inconsistent diagnostic process, that currently depends on an overstretched staff noticing distress on their own, with a fast, structured screening step. This frees staff time for the part of the process that actually requires a human: interpretation and support. Since evidence on teacher and student well-being consistently finds that principal and staff support is one of the strongest predictors of reduced burnout and stress (Martins, 2020; Carlotto, 2002), the tool is designed to strengthen that human layer, not replace it. For the same reason, aggregation and confidentiality safeguards are a foundational need. Research on school-based mental health tools warns that even well-intentioned monitoring can produce a perverse incentive, with students concealing distress for fear of being labelled if confidentiality is not technically guaranteed (Ovi et al., 2026).

3.3. Political Landscape

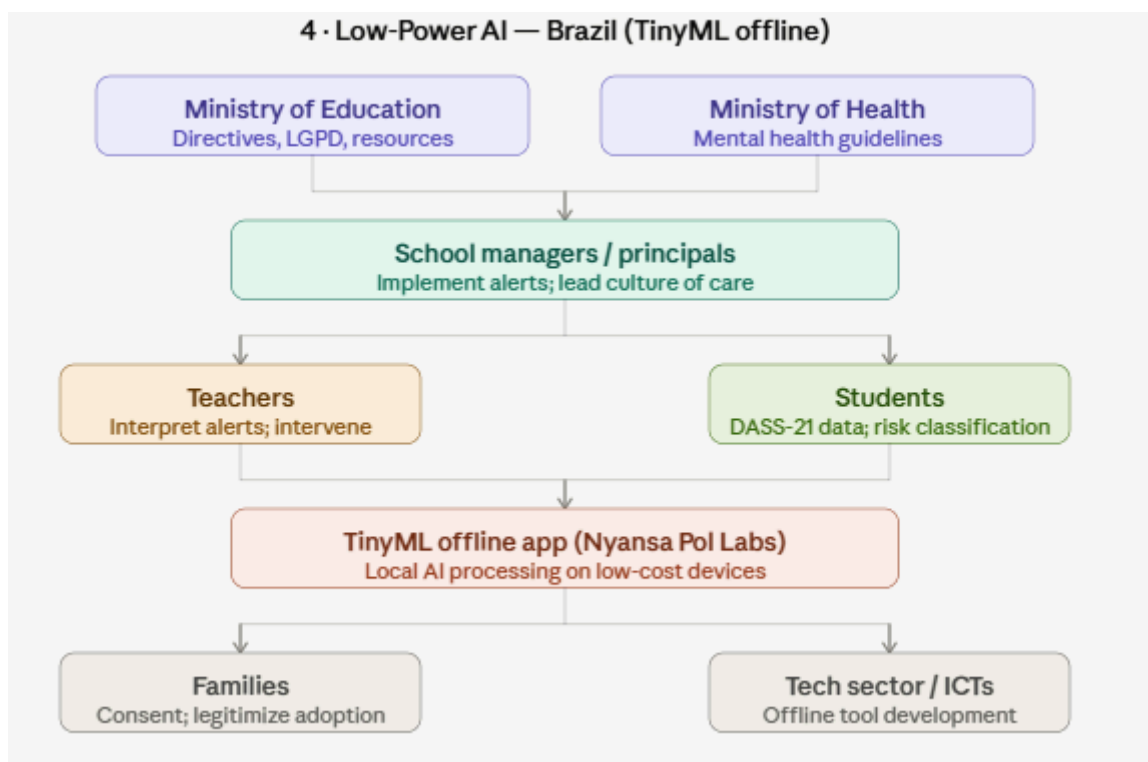
Brazil has already taken a legislative step in this direction: the National Policy of Psychosocial Attention in School Communities (Law 14.819/2024) formally recognises mental health as a component of education quality. The gap is now implementation capacity. Education, health, and social-assistance sectors remain fragmented, and chronic underfunding (an accumulated R\$117.71 billion shortfall in the Knowledge Budget since 2014; Knowledge Observatory, 2023), has left schools without scalable mechanisms to act on the policy's intent. This is a case of policy failure through insufficient instruments rather than absence of diagnosis (Lima & D'Ascenzi, 2013). At the same time, there is a relevant political opportunity: Brazil recorded a 2.3% drop in basic education enrollment in 2025 (BRAZIL, 2024), and interventions that can demonstrably improve that behaviour, offering measurable outcomes and positive political response, will likely be a priority for education authorities.

3.4. Stakeholders

The Ministries of Education and Health hold central authority, setting guidelines, allocating resources, and establishing regulatory frameworks that ensure the tool's compliance with Brazil's General Data Protection Law (BRASIL, 2018) and national school mental-health guidance (BRAZIL, 2023); their interest is high given the political return described above. School principals and teachers are responsible for day-to-day implementation and interpreting alerts, but their willingness to adopt the tool depends on it being perceived as support rather than added workload, which points directly to the need for training in emotional and instrumental response.

Students are the group most affected, and their engagement hinges on trust, particularly confidence that data will not be used to label or stigmatise them. Parents and guardians

shape the tool's legitimacy through consent and either reinforce or resist adoption depending on how privacy is communicated. Finally, technology partners and civil-society organisations, including groups such as Nyansa Pol Labs, which specialise in offline-first educational tools for the Global South (Plancher et al., 2024), provide the technical capacity to build and adapt the system to local conditions.



3.5. Challenges and Gaps

Three structural gaps intersect here. Technologically, most existing AI solutions were not designed for low-infrastructure environments (Ovi et al., 2026). Institutionally, chronic underfunding and a shortage of specialised professionals limit schools' capacity to act on what any screening tool reveals (Carlotto, 2002). Socioculturally, stigma discourages students from disclosing distress in the first place. None of these gaps is solved by better technology alone; each requires a corresponding governance and training response, which is why the recommendations below pair the technical solution with institutional and communication measures rather than presenting it in isolation.

3.6. Recommendations

We recommend adopting a public policy built around low-power, offline-first AI (TinyML) as the core mechanism for monitoring student well-being in low-connectivity contexts. This approach allows data processing to occur locally, in low-cost devices, amplifying the policy's reach to historically excluded regions. The table x.x compares TinyML to both a high

connectivity alternative and in-person counselor hiring, analysing its suitability for the Global South.

	Technically Correct	Politically Supportable	Organizationally Implementable
Offline-First Classification App	HIGH: Low-power solution resolves the connectivity challenge.	MEDIUM-HIGH: High public interest in well-being, but risk of opposition due to privacy concerns.	MEDIUM: Requires specialized partnership (TinyML) and management of teacher/staff workload.
Universal Counselor Hiring	HIGH: Direct support solution and scientifically proven.	MEDIUM: Opposition due to high recurring cost and budgetary sustainability.	LOW: Difficulty in finding and retaining qualified professionals in low-income rural areas.
High-Connectivity Platform	HIGH: High diagnostic accuracy with Big Data.	MEDIUM-LOW: High cost and political resistance to large infrastructure investments.	LOW: Lack of internet infrastructure in schools makes it unfeasible for the Global South.

This could be implemented through four coordinated actions:

a) Practical Implementation

First, pilot and scale carefully. We argue for the development of an offline-first mobile application, capable of collecting data about students' well-being from a validated psychometric instrument (DASS-21) — a 21 item questionnaire that assesses levels of depression, anxiety, and stress, identifying recent emotional state in educational and health research, allowing screening and monitoring without diagnostic function, adapted and validated for Brazilian adolescents. The application processes questionnaire data locally through lightweight machine learning models, classifying students into risk levels as an aggregated and anonymized alert system, guiding schools' action without replacing professional judgment.

Implementation should begin with around 100 schools in the Northeast and North regions, aligned with existing AIED Unplugged and TinyML pilot initiatives, with a clear, measurable target: a 15% reduction in DASS-21 scores pre- to post-intervention (Ovi et al., 2026). Thereon, pilot results can be used to validate accuracy and refine protocols before expanding nationally, which both reduces technical risk and builds the political case for scale-up.

b) Institutionalize AI-based pedagogical routines for teachers

Second, institutionalise the human response, not just the technology. Continuing-education modules should train teachers and school teams to interpret alerts and respond with emotional, informational, and instrumental support (Carlotto, 2002). Equally important is ensuring the tool does not become one more item on an already-overloaded plate: reorganising tasks, reducing unrelated administrative burden, and offering institutional incentives will determine whether staff experience the system as support or surveillance.

c) Building Monitoring with Focus on Equity and Data Transparency

Third, build trust through data governance, explicitly and visibly. Local, encrypted storage; individual-level access restricted to trained counsellors; aggregated, anonymised reporting for managers; and an explicit, communicated prohibition on using the data for punitive or discriminatory purposes are not implementation details, they are the precondition for student and family participation, given the stigma documented above.

d) Research Translation

Fourth, invest in research translation. Adapting DASS-21 for everyday school use, producing clear guidance for non-specialist staff, and tracking both quantitative indicators (DASS-21 scores, dropout rates) and qualitative signals (school-community perception) will determine if the tool is able to become a genuine, evolving instrument of care.

Taken together, these measures reframe the central claim of this case study: although low-power AI is not, by itself, a solution to student well-being in Brazil, it is a way of making an already understood and legislated problem (the absence of systematic, trusted, and timely psychosocial support) tractable in the schools that today have the least capacity to address it.